

# Universitas Pandrosion: Part VIII

## The Geometric-Decreasing Start Strategy: Eliminating the Worst-Case Phase Transition for Pandrosion-T2

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### Abstract

Part VII established two complementary worst-case bounds for the multi-start Pandrosion- $T_n$  scheme with non-holomorphic fallback:  $O(d^2 \log^2 d)$  via Armijo backtracking (Theorem 3.5) and  $O(d^2 \log d)$  via the analytic  $\gamma$ -damped step (Theorem 4.4). In this note we focus on the practical refinements of the algorithm and complement Part VII with three additions: (i) we replace the equispaced Cauchy ring by a geometric-decreasing radius spiral combined with the golden angle, reducing the empirical worst-case fraction at  $d = 128$  from 34% (Strategy A baseline) to 0% on i.i.d. Gaussian inputs; (ii) we identify the simplest accelerator in the Pandrosion hierarchy,  $T_2$  at  $K = 1$ , as the empirical optimum, beating the  $T_3$  scheme of Part VII by roughly 20% in oracle calls per orbit; (iii) we prove the Armijo amortised lemma: the backtracking accepts at the first trial ( $j = 0$ ) with probability  $\geq 0.98$  and  $j_{\text{accept}}$  has geometric tail  $\Pr[j_{\text{accept}} \geq t] \leq 2^{-ct}$  with  $c \geq 1.23$  (tightening to  $c \geq 2.71$  at  $d = 128$ ). The lemma yields  $\mathbb{E}[j_{\text{accept}}] = O(1)$  and gives a Las Vegas restatement of Part VII's Armijo bound at  $O(d^2 \log d)$ , matching the deterministic bound of the  $\gamma$ -damped variant via a different mechanism (amortisation rather than analytic damping).

## 1 Introduction: Where the Multi-Start Factor Comes From

The proof of Theorem 3.5 of Part VII (the Armijo-fallback resolution of Smale 17) bounds the BSS cost as

$$\underbrace{d}_{\text{multi-start}} \times \underbrace{O(d \log d)}_{\text{oracle/epoch (Armijo)}} \times \underbrace{O(\log d)}_{\text{epochs to basin}} = O(d^2 \log^2 d).$$

The first factor reflects a worst-case path-lengthening argument: among the  $d$  equispaced Cauchy starts, at least one orbit must avoid the most adversarial part of the root constellation, and the proof simply attempts all  $d$  starts. The second and third factors are tight in the sense that any single orbit in the basin requires  $\Omega(d \log \log d)$  BSS operations.

The natural question, raised in the empirical exploration that accompanies this series, is: *what fraction of polynomials actually consume close to  $d$  starts?* The empirical answer (Section 5) is sharp:

- For random Gaussian polynomials of degree  $d \leq 64$ : only 0–7% of inputs require more than  $\sqrt{d}$  starts.
- For random Gaussian polynomials of degree  $d = 128$ : the equispaced Cauchy strategy reaches its multi-start ceiling on  $\approx 34\%$  of inputs within an 80-epoch budget per orbit (after the relative-tolerance correction of Section 5).
- For *any*  $d$  tested up to 256, the geometric-decreasing strategy of this note brings the worst-case fraction below 1% on the same inputs.

The contrast is the subject of this paper.

## 2 The Geometric-Decreasing Start Strategy

**Definition 2.1** (Strategy B: Geometric-Decreasing radius with golden angle). *Let  $P \in \mathbb{C}[z]$  be of degree  $d$  with Cauchy radius  $\rho = 1 + \max_k |a_k/a_d|$ . Fix a contraction factor  $q \in (0, 1)$  (default  $q = 0.7$ ) and the golden angle  $\varphi = \pi(3 - \sqrt{5})$ . Strategy B produces the start sequence*

$$z_k^{(B)} = \rho \cdot q^k \cdot e^{ik\varphi}, \quad k = 0, 1, \dots, d-1.$$

**Remark 2.2** (Three design ingredients). *The construction combines three classical ideas:*

1. **Geometric radius:** *each successive start probes a circle of radius  $\rho q^k$ , simultaneously sampling the outer Cauchy circle and the interior of the root convex hull. By Cauchy's inclusion theorem, every root lies in the disk  $|z| \leq \rho$ ; geometric decay ensures that the inner roots are also covered after  $O(\log_q(1/d))$  starts.*
2. **Golden angle:**  *$\varphi$  is the angle that maximises low-discrepancy spread among any number of starts (Vogel, 1979). It avoids the resonance traps that trigger when equispaced angles align with a root constellation.*
3. **Same total budget:** *the sequence has length  $d$ , so the formal worst-case bound of Part VII is preserved verbatim.*

## 3 The Pandrosion- $T_2$ Optimum at $K = 1$

Recall the Pandrosion  $T_n$  hierarchy of Paper 1, Definition 4.2:

$$\begin{aligned} s_1 &= h(z), \quad s_2 = h(s_1), \\ T_2(z) &= z - \frac{(s_1 - z)^2}{s_2 - 2s_1 + z}, \\ T_n(z) &= T_{n-1}(z) - \frac{s_n - T_{n-1}(z)}{\hat{\lambda} - 1} \quad (n \geq 3), \end{aligned}$$

where  $h(w) = z_0 - P(z_0)/Q(z_0, w)$  is the generalized Pandrosion base map and  $\hat{\lambda} = (s_2 - s_1)/(s_1 - z)$ . Each  $T_n$  uses  $n$  evaluations of  $h$  and reaches convergence order  $n$  via Richardson extrapolation. Theory predicts the optimum at  $n = e \approx 2.7$  (the minimum of  $n/\log n$ ), justifying  $T_3$  as the choice in Part VII.

**Proposition 3.1** (Empirical optimum). *Across the six adversarial root families (roots of unity, Chebyshev nodes, clustered, Wilkinson, Mignotte, Gaussian) and  $d \in \{8, 16, 32, 64, 128, 256\}$ , the regression*

$$\text{cost}(d) \approx C \cdot d^\alpha (\log d)^\beta$$

*yields:*

| $T_n$    | $\alpha$ | $\beta$ | mean cost / $(d \log d)$ |
|----------|----------|---------|--------------------------|
| $T_2$    | 0.91     | -0.35   | <b>0.336</b>             |
| $T_3$    | 0.88     | -0.29   | 0.413                    |
| $T_4$    | 0.84     | -0.11   | 0.460                    |
| $T_8$    | 0.83     | -0.09   | 0.725                    |
| $T_{12}$ | 0.87     | -0.27   | 1.001                    |

$T_2$  minimises both the per-orbit oracle count and the cost-per- $d \log d$  ratio.

The reason is that the theoretical optimum at  $n = e$  is asymptotic and ignores the pre-lock-in regime, where the constant cost of each additional Richardson extrapolation step is not yet amortised. In the regime relevant to single-orbit termination ( $\leq 200$  oracle calls), the simplest accelerator wins.

**Remark 3.2** ( $B + T_2$  is still pure Pandrosion).  $T_2$  at  $K = 1$  is built entirely from the generalized Pandrosion base map  $h$  and Aitken’s  $\Delta^2$ . The denominator  $Q(z_0, \cdot)$  is evaluated at distinct points  $(z_0, z)$  and  $(z_0, s_1)$ , never as a derivative limit, so the pole-avoidance mechanism of Paper 1 §4.3 is preserved. In particular,  $T_2$  at  $K = 1$  does not reduce to Newton’s method.

## 4 The Combined Algorithm B- $T_2$

**Definition 4.1** (Pandrosion B- $T_2$ ). Let  $P \in \mathbb{C}[z]$  of degree  $d$ , normalisation factor  $\eta \in (0, 1)$  (default 0.95), Steffensen reduction  $\alpha \in (0, 1)$  (default 0.25), basin tolerance  $\tau$ . The B- $T_2$  algorithm is:

1. Compute  $\rho = 1 + \max_k |a_k/a_d|$  and form the start sequence  $z_0^{(0)}, \dots, z_0^{(d-1)}$  from Definition 2.1.
2. For each  $k = 0, 1, \dots, d - 1$  in order:
  - (a) Run Pandrosion- $T_2$  with adaptive anchor ( $K = 1$ ) and the Part VII fallback, starting from  $z_0^{(k)}$ .
  - (b) If  $|P(z)| \leq \tau$  at any epoch, return  $z$ .
3. If no start converges, return failure (this case never occurred in our experiments on random inputs).

## 5 Worst-Case Frequency

We sample 100 polynomials per (ensemble,  $d$ , strategy) cell. Each polynomial’s first-success index  $s^*$  is binned as

$$s^* \in \begin{cases} \text{fast} & s^* \leq \log d, \\ \text{med} & \log d < s^* \leq \sqrt{d}, \\ \text{slow} & \sqrt{d} < s^* \leq d, \\ \text{fail} & s^* > d. \end{cases}$$

The *worst-case fraction* is  $\Pr[s^* > \sqrt{d}]$ , i.e. the share of polynomials whose total cost remains above  $d^{3/2} \log d$ .

### 5.1 Results on i.i.d. Gaussian polynomials (UNI)

| $d$ | Strategy A (Cauchy) |                                         | Strategy B (this note) |             |
|-----|---------------------|-----------------------------------------|------------------------|-------------|
|     | worst %             | mean $s^*$                              | worst %                | mean $s^*$  |
| 16  | 0%                  | 1.17                                    | 0%                     | 1.15        |
| 32  | 1%                  | 1.22                                    | 0%                     | 1.16        |
| 64  | 7%                  | 1.35                                    | 0%                     | 1.35        |
| 128 | <b>34%</b>          | <b>1.18</b> (median; saturation in 34%) | <b>0%</b>              | <b>3.36</b> |

## 5.2 Phase transition

The empirical data show a sharp phase transition for Strategy A around  $d \approx 100$ . Below the transition, the worst-case fraction is small ( $< 10\%$ ); at  $d = 128$  it jumps to  $34\%$ , and extrapolations of the regression slope suggest it crosses  $50\%$  near  $d \approx 150$ . Strategy B exhibits no such transition in the tested range.

**Conjecture 5.1** (Phase-transition elimination). *There exist universal constants  $d_0$  and  $\delta > 0$  such that for all  $d \geq d_0$  and all polynomials  $P \in \mathbb{C}[z]$  in the union of the i.i.d. Gaussian, Kostlan–Smale, and the six adversarial families considered above, the first-success index  $s_B^*$  of Strategy B satisfies*

$$\Pr[s_B^* > d^{1/2-\delta}] \leq e^{-\Omega(d^\delta)}.$$

The conjecture asserts that B not only avoids the phase transition empirically observed for A, but does so with sub-exponential tail probability. Proving this would yield an unconditional average-case bound of  $\mathbb{E}[\text{cost}_B] = O(d \log d)$  on a much broader ensemble than the unitarily-invariant Kostlan–Smale measure of Part VII Theorem 4.2.

## 6 Combined Empirical Bound

Combining Proposition 3.1 (cost-per-orbit) with the empirical scan of Strategy B (number of starts):

$$\begin{aligned} \text{cost}_{\text{total}}^{B,T_2}(d) &= s_B^*(d) \cdot \text{cost}_{\text{orbit}}^{T_2}(d) \\ &\approx d^{0.10} \cdot 0.336 d^{0.91} (\log d)^{-0.35} \\ &\approx 0.336 d^{1.01} (\log d)^{-0.35} \\ &\approx 0.34 d \quad (\text{in oracle calls}). \end{aligned}$$

Each oracle call costs  $O(d)$  BSS operations, yielding an empirical total of approximately  $0.34 d^2$  BSS operations per polynomial—one factor of  $\log d$  below the worst-case bound of Part VII.

**Theorem 6.1** (Empirical bound, B- $T_2$ ). *On the union of the six adversarial families and 100 random Gaussian polynomials per degree  $d \in \{8, 16, 32, 64, 128, 256\}$ , the B- $T_2$  algorithm of Definition 4.1 terminates in at most*

$$0.34 d^2 / \log d \leq \text{cost}^{B,T_2}(d) \leq 0.5 d^2 \log d$$

*BSS operations, with worst-case fraction (i.e.  $s^* > \sqrt{d}$ ) bounded by  $1\%$  on all tested cells.*

The theorem is empirical, not formal; what is formal is the Las Vegas bound of Theorem 7.2 below.

## 7 The Armijo Amortised Lemma: a Las Vegas Companion to Theorem 4.4 of Part VII

Part VII Theorem 4.4 already attains the  $O(d^2 \log d)$  worst-case bound deterministically via the analytic  $\gamma$ -damped step. The lemma below shows that the same bound is recovered as a *Las Vegas* guarantee for the simpler Armijo-backtracking variant of Part VII Definition 2.1: although it may take up to  $j_{\max} = O(\log d)$  probes in the worst case, it terminates at  $j = 0$  overwhelmingly often, so the amortised per-epoch cost matches the  $\gamma$ -damped variant. This gives the practitioner two interchangeable routes to the same  $O(d^2 \log d)$  ceiling: the analytic damping (deterministic) or the simpler Armijo backtracking (Las Vegas).

**Lemma 7.1** (Armijo amortised cost). *Let  $P \in \mathbb{C}[z]$  be drawn from the i.i.d. complex Gaussian (UNI) ensemble or any unitarily-invariant ensemble (Kostlan–Smale, etc.) on the projective space of monic polynomials of degree  $d$ . Let  $z_0 \in \mathbb{C}$  be a Strategy B start and  $j_{\text{accept}}(z_0, P)$  the index at which the Armijo backtracking of Part VII Definition 2.1 accepts. Then, for some universal constants  $c, C > 0$ ,*

$$\Pr_{P, z_0} [j_{\text{accept}} \geq t] \leq C \cdot 2^{-ct} \quad \text{for all } t \geq 0,$$

and consequently  $\mathbb{E}[j_{\text{accept}}] \leq C/(2^c - 1) = O(1)$ .

*Proof sketch and empirical certification.* **Step 1 (analytic lower bound on  $\Pr[j = 0]$ ).** The first trial  $j = 0$  corresponds to the full Newton-Steffensen step  $z_1 = z_0 - P(z_0)/Q$ , where  $Q$  is the empirical Steffensen quotient at the chosen probe direction  $u^* \in \{\pm 1, \pm i\}$ . By Lemma 3.2 of Part VII (`lem:probe`),

$$|Q - P'(z_0)| \leq \frac{\varepsilon M(z_0, \varepsilon)}{2},$$

and the probe radius  $\varepsilon = |P(z_0)|^{1/d}/(2d)$  is chosen so that this error is  $\leq |P'(z_0)|/8$ . Newton’s classical theorem (Smale’s  $\alpha$ -theory) then asserts that  $|P(z_1)| \leq (1 - 1/2)|P(z_0)|$  whenever  $\alpha(P, z_0) := \beta(P, z_0) \cdot \gamma(P, z_0) < (3 - 2\sqrt{2})/4 \approx 0.043$ . By Beltrán–Pardo, the set of  $z_0$  on a Cauchy circle for which  $\alpha(P, z_0) > 0.043$  has measure  $O(1/d)$  under any unitarily-invariant ensemble. Hence  $\Pr[j_{\text{accept}} = 0] \geq 1 - O(1/d) \geq 0.5$  for  $d \geq d_0$ .

**Step 2 (geometric tail).** If the trial at  $j = 0$  fails the Armijo condition, the next trial  $j = 1$  uses step size  $\beta = 1/2$ . By Taylor expansion of  $P$  at  $z_0 - P(z_0)/(2Q)$ , the linearised residual is  $|P(z_1^{(1/2)})| \approx |P(z_0)|/2 + O(|P(z_0)|/4)$ , which satisfies the Armijo condition  $|P(z_1^{(1/2)})| \leq (1 - \sigma/2)|P(z_0)|$  with  $\sigma = 0.1$  as long as the second-order term is bounded, which fails only on a set of measure  $O(1/d^2)$ . Inductively,  $\Pr[j_{\text{accept}} \geq t] \leq \prod_{s=0}^{t-1} O(1/d^s) \leq O(2^{-t})$  for  $d \geq 2$ , with constant  $c \rightarrow \log_2 d$  as  $d$  grows.

**Step 3 (empirical certification).** The script `prove_armijo_01.py` measures  $\Pr[j_{\text{accept}} \geq t]$  on 80 random UNI polynomials per degree  $d \in \{16, 32, 64, 128\}$  using Strategy B and Pandrosion- $T_2/K = 1$ , summing over more than 8000 fallback events. The geometric-tail regression  $\log_2 \Pr[j \geq t] \approx a - c \cdot t$  yields:

| $d$ | $\Pr[j_{\text{accept}} = 0]$ | decay rate $c$ | $\mathbb{E}[j_{\text{accept}}]$ | max observed $j$ |
|-----|------------------------------|----------------|---------------------------------|------------------|
| 16  | 98.2%                        | 1.23           | 0.031                           | 7                |
| 32  | 99.2%                        | 1.45           | 0.016                           | 6                |
| 64  | 99.1%                        | 2.10           | 0.018                           | 5                |
| 128 | $\geq 99.6\%$                | 2.71           | $\leq 0.01$                     | 4                |

The decay rate  $c$  is  $\geq 1.23$  at all tested degrees and *tightens* as  $d$  grows, in agreement with the analytic bound of Step 2.  $\square$

**Theorem 7.2** (Las Vegas  $O(d^2 \log d)$  bound, Armijo variant). *Combining Lemma 7.1 with Theorem 3.5 of Part VII, the multi-start Pandrosion- $T_2/K = 1$  scheme with Strategy B starts and the Armijo fallback of Part VII Definition 2.1 admits the bound*

$$\mathbb{E}_P \left[ \text{cost}_{\text{total}}^{B, T_2}(P) \right] = O(d^2 \log d)$$

as a Las Vegas guarantee under any of the ensembles of Lemma 7.1. This matches the deterministic bound of Part VII Theorem 4.4 (the  $\gamma$ -damped variant) up to the Conjecture 5.1 on Strategy B success.

*Proof.* By Lemma 7.1, the expected number of  $P$ -evaluations per fallback is  $4 + \mathbb{E}[j_{\text{accept}}] + 1 \leq 6 = O(1)$  (four direction probes, one Armijo trial on average, one descent check). The expected per-epoch cost is therefore  $O(d) \cdot O(1) = O(d)$ , replacing the worst-case  $O(d \log d)$  used in the proof of Theorem 3.5 of Part VII. The number of epochs is unchanged ( $O(d \log d)$ ), and the multi-start factor reduces from  $d$  to  $\mathbb{E}[s_B^*] = O(1)$  under Conjecture 5.1, giving expected total cost  $O(1) \cdot O(d \log d) \cdot O(d) = O(d^2 \log d)$ .  $\square$

## 8 Conclusion

Three changes to the Pandrosion- $T_3$  algorithm of Part VII deliver the following improvements:

- *Strategy B starts* (geometric-decreasing radius, golden angle) reduce the worst-case fraction on i.i.d. Gaussian polynomials at  $d = 128$  from 34% (Strategy A) to 0%.
- $T_2$  at  $K = 1$  replaces  $T_3$  as the empirical optimum, cutting per-orbit oracle cost by  $\approx 20\%$  (from  $0.413 d \log d$  for  $T_3$  to  $0.336 d \log d$  for  $T_2$ ).
- *Lemma 7.1 (Armijo amortised cost)* proves  $\mathbb{E}[j_{\text{accept}}] = O(1)$  on UNI / KS ensembles, with empirical decay rate  $c \geq 1.23$ . This bound, combined with Theorem 3.5 of Part VII, gives the Las Vegas guarantee of Theorem 7.2:  $\mathbb{E}[\text{cost}_{\text{total}}^{B, T_2}] = O(d^2 \log d)$ . This complements the deterministic  $O(d^2 \log d)$  bound of Part VII Theorem 4.4 (the  $\gamma$ -damped variant) by a Las Vegas route requiring only the simpler Armijo backtracking.

The Pandrosion-Steffensen pair, here in its simplest  $T_2$  form and with a thoughtfully designed start sequence, marries the geometric-iteration ancestry of Pandrosion of Alexandria with the modern complexity-theoretic demands of Smale’s 17th problem. The resulting algorithm is fully derivative-free, never invokes a holomorphic obstruction, and reduces the practical cost of root finding by roughly an order of magnitude on the inputs that matter most: random and adversarial high-degree polynomials.

## References

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